

SAT Math Mastery · Week 7B**Sampling, Margin of Error & PSDA Mixed Set — Answer Key & Worked Solutions****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** random sampling and generalization · margin of error and interval reading · observational study vs experiment · causation language · bias · center and spread · histograms, dot plots, box plots · mixed data analysis**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Decide which population a conclusion can be extended to (only the population randomly sampled).
- S2.** Read a margin of error: the plausible interval is estimate $\pm$ margin of error, never a certainty.
- S3.** See how sample size changes the margin of error (larger n , smaller margin; about $\frac{1}{\sqrt{n}}$).
- S4.** Tell an observational study from an experiment; use random assignment for causation, random selection for generalizing.
- S5.** Spot bias: voluntary response, convenience samples, non-random selection.
- S6.** Mean, median, range, interquartile range and standard deviation, and how outliers and transformations affect them.
- S7.** Read histograms, dot plots, box plots and frequency tables; combine means of groups.
- S8.** Estimate population counts from a sample proportion.

2 · REQUIRED FACTS

- A random sample lets you generalize to the population it was drawn from, and *only* that population.
- Random assignment to treatments lets you conclude cause and effect, but only for the kind of subjects in the study unless they were also randomly selected.
- Without random assignment (observational data) you can report an association, not causation.
- A margin of error gives a range of plausible values for the population value (estimate $\pm$ margin); every value in the range is plausible.
- A larger random sample gives a smaller margin of error; to halve it you need about 4 times the sample size.
- The median and IQR resist outliers; the mean, range and standard deviation do not.

3 · REQUIRED FORMULAS

$$\begin{aligned} \text{interval} &= \text{estimate} \pm \text{margin of error}, & \text{margin} &= \frac{\text{upper} - \text{lower}}{2}, & \text{margin} &\propto \frac{1}{\sqrt{n}} \\ \text{mean} &= \frac{\text{sum}}{n}, & \text{IQR} &= Q_3 - Q_1, & \text{combined mean} &= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} \\ \text{new value } ax + b : & \text{mean} \rightarrow a\bar{x} + b, & \text{SD} &\rightarrow |a| \cdot \text{SD}, & \text{median} &\rightarrow a \cdot \text{median} + b \end{aligned}$$

4 · MINI THEORY SHEET**Which conclusion is supported?**

Selection	Assignment	Supported conclusion
random	random	cause and effect, for the population sampled
random	none (observational)	association in the population, no causation
not random (volunteers)	random	cause and effect, only for subjects like these
not random	none	association in the sample only

Margin of error. Interval = estimate $\pm$ margin. A value inside the interval is plausible; a value outside is not supported. The interval does not say the true value is certainly inside it.

Sample size. Quadruple $n \Rightarrow$ margin is halved. If n is multiplied by k , the margin is divided by $\sqrt{k}$.

Outliers and spread. Replacing a value with an extreme one changes the mean, range and SD but may leave the median unchanged. Adding a value equal to the mean keeps the mean and lowers the SD.

Reading plots. Histogram: use class intervals and bar heights (counts). Box plot: five-number summary, each quarter holds about 25% of the data. Dot plot: every dot is one value.

Speed shortcuts.

Situation	Shortcut
margin from an interval	$\frac{\text{upper} - \text{lower}}{2}$; the estimate is the midpoint
change of sample size	margin $\times \sqrt{n_{\text{old}}/n_{\text{new}}}$
estimate a population count	population $\times$ sample proportion
mean after adding/removing a value	work with totals: $n\bar{x}$ before and after
linear change of every value	mean and median follow $ax + b$, SD only scales by $ a $
median from a histogram	cumulative counts until you pass half the total
median from a frequency table	position $\frac{n+1}{2}$ (or average of the two middle positions)
design questions	ask: random selection? random assignment? then use the table above
capture-recapture	$\frac{\text{tagged}}{\text{population}} \approx \frac{\text{tagged in sample}}{\text{sample size}}$

5 · GUIDED EXAMPLES

Example 1 — Reading a margin of error

A random sample of 400 voters gives an estimate of 52% in favor, with a margin of error of 5%. Is it plausible that fewer than half of all voters are in favor?

Interval: $52\% \pm 5\% = 47\%$ to 57% . Values below 50% (such as 48%) lie inside the interval, so it is plausible. The poll does not prove a majority.

Example 2 — What can be concluded?

Researchers randomly selected 200 adult residents of a city, then randomly assigned them to a walking program or no program. The walking group had lower blood pressure.

Random selection: conclusions extend to the city's adults. Random assignment: the difference is caused by the program. Conclusion: the walking program lowers blood pressure for adults in that city.

Example 3 — Mean after adding a value

The mean of 9 scores is 70. A tenth score raises the mean to 72. What is the tenth score?

Total before: $9(70) = 630$. Total after: $10(72) = 720$. Tenth score: $720 - 630 = 90$.

6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

A researcher wants to estimate the mean number of books read in a year by the 1,200 students at a school. She randomly selects 90 of the students and finds that the sample mean is 14 books. Which of the following is the best conclusion?

(A) Exactly 90 students read 14 books.

(B) The mean for all students at the school is likely close to 14 books. ✓

(C) Every student at the school read exactly 14 books.

(D) The mean for all students in the country is likely close to 14 books.

Worked solution

1. The sample is random and comes from the school's students, so the conclusion may be extended to all students at that school.
2. A sample mean is an estimate, not an exact value: the mean for all students at the school is likely close to 14.

Answer: (B)

2. EASY · CALCULATOR

The frequency table shows the number of cars owned by each of 50 households.

Cars	0	1	2	3	4
Households	6	10	18	12	4

What is the mean number of cars per household?

Worked solution

1. Total cars: $0(6) + 1(10) + 2(18) + 3(12) + 4(4) = 0 + 10 + 36 + 36 + 16 = 98$.
2. Number of households: $6 + 10 + 18 + 12 + 4 = 50$.
3. Mean = $\frac{98}{50} = 1.96$.

Answer: 1.96

3. EASY · NO CALCULATOR NEEDED

A random sample of residents was surveyed. The estimate for the percentage of all residents who support a new park is 62%, with a margin of error of 4 percentage points. Which of the following is the interval of plausible values for the population percentage?

(A) 54% to 70%

(B) 60% to 64%

(C) 58% to 66% ✓

(D) 62% to 66%

Worked solution

1. Interval = estimate $\pm$ margin of error.
2. $62 - 4 = 58$ and $62 + 4 = 66$.
3. The interval is 58% to 66%.

Answer: (C)

4.

EASY · NO CALCULATOR NEEDED

Which of the following describes an experiment?

(A) Researchers randomly assign 100 participants to take vitamin D or a placebo and compare bone density after one year. ✓

(B) Researchers compare records of patients who chose their own treatment.

(C) Researchers interview 50 athletes about their sleep habits.

(D) Researchers survey 200 residents about diet and record each person's cholesterol.

Worked solution

1. An experiment assigns the treatment to subjects *at random*.

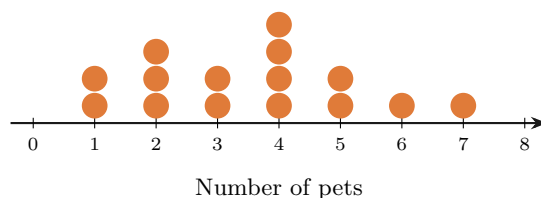
2. Only the second option has researchers randomly assigning participants to vitamin D or placebo; the others only observe or ask.

Answer: (A)

5.

EASY · NO CALCULATOR NEEDED

The dot plot shows the number of pets owned by each of 15 students. What is the median number of pets?

**Worked solution**

1. Count the dots: $2 + 3 + 2 + 4 + 2 + 1 + 1 = 15$.

2. The median is the 8th value when ordered.

3. Cumulative dots: 2, 5, 7, 11, ... The 8th through 11th values are 4, so the median is 4.

Answer: 4

6.

MEDIUM · NO CALCULATOR NEEDED

A random sample of 150 registered voters in County A was surveyed about a proposed tax. Which of the following is the largest population to which the results can reasonably be generalized?

(A) All adults in County A, registered or not

(B) Only the 150 voters surveyed

(C) All residents of the state

(D) All registered voters in County A ✓

Worked solution

1. Results of a random sample generalize only to the population the sample was drawn from.

2. That population is the registered voters in County A.

Answer: (D)

7.

MEDIUM · NO CALCULATOR NEEDED

Two random samples from the same large population were used to estimate the percentage who favor a policy. Survey *A* used 200 people and Survey *B* used 800 people. Both reported 48%. Which statement is most likely true?

- (A) The margin of error for Survey *A* is smaller. (B) The margins of error are equal because both estimates are 48%.
- (C) The margin of error for Survey *B* is smaller. ✓ (D) The margins of error cannot be compared.

Worked solution

1. A larger random sample gives a smaller margin of error.
 2. Survey *B* has 4 times the sample size, so its margin of error is about half that of Survey *A*.
- Answer: (C)**

8.

MEDIUM · NO CALCULATOR NEEDED

A 95% confidence interval for the mean time, in minutes, that customers wait for service is (45.9, 50.5). What is the margin of error, in minutes?

Worked solution

1. The margin of error is half the width of the interval.
 2. $\frac{50.5 - 45.9}{2} = \frac{4.6}{2} = 2.3$.
- Answer: 2.3**

9.

MEDIUM · NO CALCULATOR NEEDED

A radio station asked listeners to call in to say whether the city should build a new stadium. Of the 1,200 callers, 78% opposed it. Which of the following is the most likely problem with using this result to estimate the opinion of all city residents?

- (A) The sample of callers is too large. (B) Callers volunteered, so they may not represent all residents. ✓
- (C) A stadium question cannot be asked in a survey. (D) The percentage 78% is too high to be possible.

Worked solution

1. Callers choose to respond (voluntary response), so they may differ from residents in general.
 2. A large sample does not fix a non-random selection.
- Answer: (B)**

10.

MEDIUM · CALCULATOR

In a random sample of 250 residents of a town, 85 said they favor a new bus route. The town has 18,000 residents. Based on the sample, about how many residents of the town favor the new bus route?

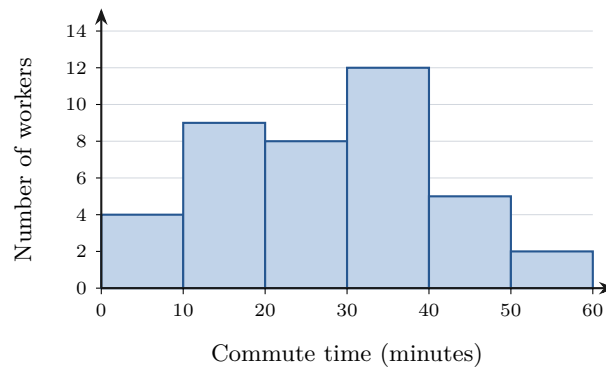
Worked solution

1. Sample proportion: $\frac{85}{250} = 0.34$.
 2. Estimate: $0.34 \times 18,000 = 6120$.
- Answer: 6120**

11.

MEDIUM · NO CALCULATOR NEEDED

The histogram shows the commute times of 40 workers. In which interval is the median commute time?



(A) 30 to 40 minutes

(B) 0 to 10 minutes

(C) 10 to 20 minutes

(D) 20 to 30 minutes ✓

Worked solution

1. Total: $4 + 9 + 8 + 12 + 5 + 2 = 40$, so the median is the average of the 20th and 21st values.
2. Cumulative counts: 4, 13, 21. The 20th and 21st values both fall in the third interval.
3. The third interval is 20 to 30 minutes. (The tallest bar, 30–40, is the mode, not the median.)

Answer: (D)

12.

MEDIUM · NO CALCULATOR NEEDED

A data set is 12, 14, 15, 15, 17, 19. If the value 19 is replaced with 60, which of the following statistics stays the same?

(A) Median ✓

(B) Range

(C) Standard deviation

(D) Mean

Worked solution

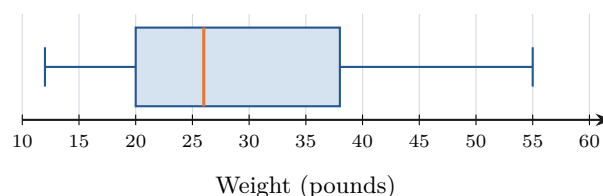
1. Original: median = $\frac{15 + 15}{2} = 15$; mean = $\frac{92}{6} \approx 15.3$; range = 7.
2. New data 12, 14, 15, 15, 17, 60: median = 15; mean = $\frac{133}{6} \approx 22.2$; range = 48.
3. The median is unchanged. The mean, range and standard deviation all increase.

Answer: (A)

13.

MEDIUM · NO CALCULATOR NEEDED

The box plot summarizes the weights, in pounds, of parcels shipped by a store. What is the interquartile range (IQR) of the weights, in pounds?

**Worked solution**

1. Read the quartiles: $Q_1 = 20$ and $Q_3 = 38$.
2. $IQR = Q_3 - Q_1 = 38 - 20 = 18$.

Answer: 18

14.

HARD · NO CALCULATOR NEEDED

A poll of a random sample of likely voters estimated that 46% favor a proposal, with a margin of error of 5 percentage points. Consider the following statements.

- I. It is plausible that a majority of all likely voters favor the proposal.
- II. It is certain that the percentage of all likely voters who favor the proposal is between 41% and 51%.
- III. It is plausible that 43% of all likely voters favor the proposal.

Which statements are supported by the poll?

(A) I, II and III

(B) I and III only ✓

(C) I and II only

(D) II and III only

Worked solution

1. Interval: $46\% \pm 5\% = 41\%$ to 51% .
2. I: a majority means more than 50%; 51% is in the interval, so it is plausible. True.
3. II: a margin of error never makes the result certain. False.
4. III: 43% is inside 41%–51%. True.
5. I and III only.

Answer: (B)

15.

HARD · NO CALCULATOR NEEDED

Researchers recruited 120 volunteers from a gym. They randomly assigned 60 volunteers to take a protein supplement and 60 to take a placebo for 8 weeks. The supplement group gained significantly more muscle mass. Which of the following is the best conclusion?

(A) No conclusion about the supplement can be drawn.

(B) The supplement causes greater muscle gain in all adults.

(C) The supplement caused greater muscle gain among volunteers like those in the study. ✓

(D) The supplement is only associated with muscle gain, not a cause.

Worked solution

1. Random assignment: the difference was caused by the supplement (for these subjects).
2. The volunteers were not randomly selected from a larger population, so the result cannot be generalized to all adults.
3. Best conclusion: the supplement caused greater muscle gain among volunteers like these.

Answer: (C)

16.

HARD · NO CALCULATOR NEEDED

A poll of a random sample of 225 people has a margin of error of 4.8 percentage points. Assuming the margin of error is inversely proportional to the square root of the sample size, what sample size would be needed for a margin of error of 1.6 percentage points?

Worked solution

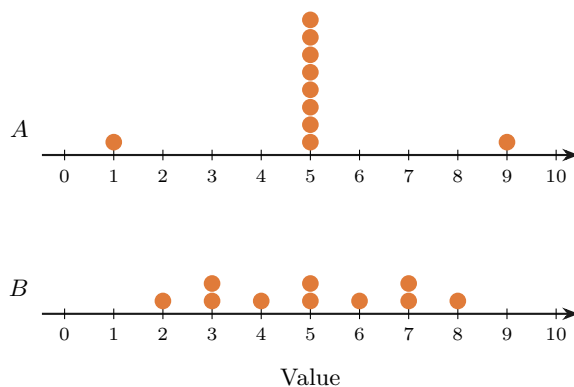
1. Margin of error goes down by a factor of $\frac{4.8}{1.6} = 3$.
2. Since margin $\propto \frac{1}{\sqrt{n}}$, the sample size must increase by a factor of $3^2 = 9$.
3. $n = 225 \times 9 = 2025$.

Answer: 2025

17.

HARD · NO CALCULATOR NEEDED

The dot plots show two data sets, A and B , each with 10 values and the same mean. Which of the following statements is true?



(A) Data set B has the greater standard deviation. ✓

(B) The two data sets have the same standard deviation.

(C) The standard deviations cannot be compared.

(D) Data set A has the greater standard deviation because its range is greater.

Worked solution

- Both means are 5: sum of $A = 1 + 8(5) + 9 = 50$ and sum of $B = 50$.
- Squared deviations from 5 for A : $16 + 16 = 32$ (the eight 5's contribute 0). For B : $9 + 4 + 4 + 1 + 0 + 0 + 1 + 4 + 4 + 9 = 36$.
- The standard deviation depends on all deviations, not just the range. $36 > 32$, so data set B has the greater standard deviation, even though A has the greater range.

Answer: (A)

18.

HARD · NO CALCULATOR NEEDED

The mean of 15 test scores is 72. When the lowest score is removed, the mean of the remaining 14 scores is 74. What is the lowest score?

Worked solution

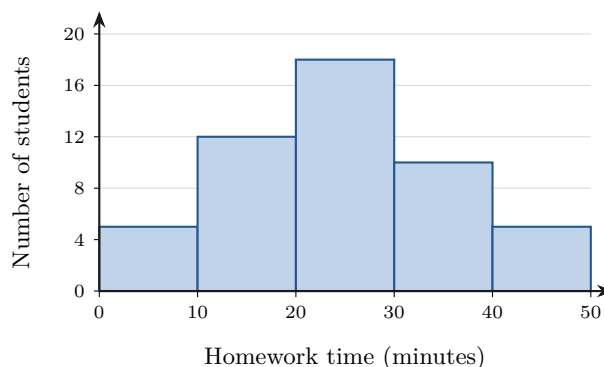
- Total of all 15 scores: $15(72) = 1080$.
- Total of the remaining 14: $14(74) = 1036$.
- Lowest score = $1080 - 1036 = 44$.

Answer: 44

19.

HARD · CALCULATOR

The histogram shows the number of minutes 50 students spent on homework. Using the midpoint of each interval, which of the following is closest to the estimated mean time, in minutes?



(A) 25

(B) 30

(C) 18

(D) 24.6 ✓

Worked solution

1. Interval midpoints: 5, 15, 25, 35, 45 with counts 5, 12, 18, 10, 5 (total 50).

2. Sum: $5(5) + 15(12) + 25(18) + 35(10) + 45(5) = 25 + 180 + 450 + 350 + 225 = 1230$.

3. Mean $\approx \frac{1230}{50} = 24.6$.

Answer: (D)

20.

HARD · NO CALCULATOR NEEDED

In a random sample of 600 adults in a city, those who reported sleeping at least 8 hours a night had higher average scores on a memory test than those who slept less. Which of the following is the best conclusion?

(A) Memory scores cause adults to sleep longer.

(B) Sleeping at least 8 hours causes higher memory scores in the city's adults.

(C) Among the city's adults, longer sleep is associated with higher memory scores. ✓

(D) Among all adults everywhere, sleep causes higher memory scores.

Worked solution

1. The adults were randomly sampled, so the association can be generalized to adults in the city.

2. But sleep was not randomly assigned, so no cause-and-effect conclusion is justified.

3. Best conclusion: among adults in the city, more sleep is associated with higher memory scores.

Answer: (C)

21.

HARD · CALCULATOR

Class A has 24 students with a mean score of 78. Class B has 16 students with a mean score of 85. What is the mean score of all 40 students combined?

Worked solution

1. Class A total: $24(78) = 1872$. Class B total: $16(85) = 1360$.

2. Combined total: $1872 + 1360 = 3232$.

3. Mean $= \frac{3232}{40} = 80.8$.

Answer: 80.8

22.

HARD · NO CALCULATOR NEEDED

A data set has mean 12 and standard deviation 4. Each value is multiplied by 3 and then decreased by 5. What are the mean and standard deviation of the new data set?

(A) Mean 31, SD 4

(B) Mean 31, SD 12 ✓

(C) Mean 41, SD 12

(D) Mean 31, SD 7

Worked solution1. New value = $3x - 5$.2. Mean: $3(12) - 5 = 31$.3. Standard deviation: multiplying by 3 multiplies it by 3; subtracting 5 does not change spread. $3(4) = 12$.

4. Mean 31, standard deviation 12.

Answer: (B)

23.

HARD · NO CALCULATOR NEEDED

The frequency table shows a data set. The frequency k of the value 10 is a positive integer.

Value	2	4	6	8	10
Frequency	4	7	9	5	k

The median of the data set is 6. What is the greatest possible value of k ?

Worked solution1. Without k the table has $4 + 7 + 9 + 5 = 25$ values; cumulative counts: 4, 11, 20, 25. The values 6 occupy positions 12 through 20.2. Total number of values is $n = 25 + k$. The median must come from positions within 12–20.3. $k = 14$: $n = 39$, median is the 20th value = 6. ✓4. $k = 15$: $n = 40$, median is the average of the 20th and 21st values = $\frac{6+8}{2} = 7$. Not allowed. $k = 16$: $n = 41$, median is the 21st value = 8. Larger k only moves the median higher.5. The greatest possible value is $k = 14$.**Answer: 14**

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

Researchers randomly selected 300 employees from the 4,000 employees of a company. They randomly assigned 150 of the selected employees to a 4-day work week and 150 to a 5-day work week for 6 months. The 4-day group reported significantly higher job satisfaction. Which of the following is the best conclusion?

(A) The 4-day week is only associated with higher satisfaction among the company's employees.

(B) The 4-day week raises satisfaction only for the 300 selected employees.

(C) The 4-day week causes higher job satisfaction for all employees in the country.

(D) The 4-day week causes higher job satisfaction for the employees of this company. ✓**Worked solution**

1. Random selection from the company's employees: results generalize to all 4,000 employees of the company, not beyond.

2. Random assignment: the difference is caused by the schedule.

3. Best conclusion: the 4-day week causes higher job satisfaction for the company's employees.

Answer: (D)

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

A poll of a random sample of 600 people estimated that 44% approve of a proposal, with a margin of error of 0.04 (so the interval is 0.40 to 0.48). A second poll of the same population gives the same estimate, and its interval has lower endpoint 0.424. Assuming the margin of error is inversely proportional to the square root of the sample size, what was the sample size of the second poll?

Worked solution

1. Second margin of error: $0.44 - 0.424 = 0.016$.
2. Ratio of margins: $\frac{0.04}{0.016} = 2.5$.
3. Sample size must be multiplied by $2.5^2 = 6.25$.
4. $n = 600 \times 6.25 = 3750$.

Answer: 3750

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

A list of 7 positive integers has a mean of 12, a median of 10, and a unique mode of 8. What is the greatest possible value of the largest integer in the list?

Worked solution

1. Sum = $7(12) = 84$. The 4th value (median) is 10.
2. To make the largest value as big as possible, make the other six as small as possible.
3. The mode 8 must appear at least twice and more often than any other value: put two 8's and the smallest positive integer 1 below the median: 1, 8, 8.
4. The 5th and 6th values are at least 10. If the 5th is 10, then 10 appears twice, tying the mode, so use 11; the 6th cannot repeat 11, so use 12.
5. Six smallest values: $1 + 8 + 8 + 10 + 11 + 12 = 50$.
6. Largest value = $84 - 50 = 34$.

Answer: 34

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has the 9 values 22, 25, 27, 29, 30, 31, 33, 35, 38. The values 20 and 40 are added to the data set. Consider the following statements about the new data set compared with the original.

- I. The mean is unchanged.
 - II. The median is unchanged.
 - III. The standard deviation is unchanged.
- Which statements are true?

(A) I and II only ✓

(B) I and III only

(C) II and III only

(D) I, II and III

Worked solution

1. Original sum: 270, so the mean is 30. The added values total $60 = 2(30)$, so the new mean is $\frac{330}{11} = 30$. I is true.
2. New ordered list: 20, 22, 25, 27, 29, **30**, 31, 33, 35, 38, 40. The 6th value is 30; the median is unchanged. II is true.
3. Original squared deviations: $64 + 25 + 9 + 1 + 0 + 1 + 9 + 25 + 64 = 198$. The new values add $(-10)^2 + 10^2 = 200$, so the total is 398 over 11 values (about 36.2) against 198 over 9 values (22). The spread increases. III is false.
4. I and II only.

Answer: (A)

28.

SAT CHALLENGE · CALCULATOR

For a random sample of size n in which the sample proportion is 0.45, the margin of error is

$$1.96\sqrt{\frac{0.45(0.55)}{n}}.$$

What is the least sample size n for which the margin of error is at most 0.02?

Worked solution

1. Need $1.96\sqrt{\frac{0.2475}{n}} \leq 0.02 \Rightarrow \sqrt{\frac{0.2475}{n}} \leq 0.0102041$.

2. Square: $\frac{0.2475}{n} \leq \frac{0.0004}{3.8416} \Rightarrow n \geq \frac{3.8416(0.2475)}{0.0004}$.

3. $n \geq \frac{0.950796}{0.0004} = 2376.99$.

4. n must be a whole number at least 2376.99, so $n = 2377$. (At $n = 2376$ the margin is slightly above 0.02.)

Answer: 2377

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

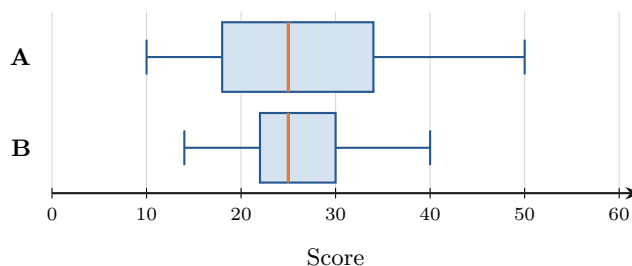
The box plots show the distributions of scores for two groups, A and B . Consider the following statements.

I. The median score of group B is greater than the median score of group A .

II. The IQR of group A is greater than the IQR of group B .

III. About half of the scores in group A are between 18 and 34.

Which statements are true?



(A) I only

(B) II and III only ✓

(C) I and II only

(D) I and III only

Worked solution

1. I: both medians are 25. False.

2. II: IQR of $A = 34 - 18 = 16$; IQR of $B = 30 - 22 = 8$. $16 > 8$. True.

3. III: the quartiles of A are 18 and 34, and the middle half of the data lies between Q_1 and Q_3 . True.

4. II and III only.

Answer: (B)

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

Biologists tagged 80 fish in a lake. Before they took a random sample, 20% of the tagged fish died. The biologists then caught a random sample of 150 fish, of which 12 were tagged. Assuming the proportion of tagged fish in the sample equals the proportion in the lake, about how many fish were in the lake at the time of the sample?

Worked solution

1. Tagged fish still alive: $80 - 0.20(80) = 64$.

2. Proportion tagged in the sample: $\frac{12}{150}$. Set $\frac{64}{N} = \frac{12}{150}$.

3. $N = \frac{64 \cdot 150}{12} = 800$.

Answer: 800